Multivariate Data

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When working with multivariate data, one common statistic people use to analyse the relationship between two features is *correlation*. Although useful in many ways, relying sorely on correlation can be dangerous and can lead to erroneous conclusions. In this week's notebook, we will go through some of the most common mistakes researchers make when analysing their data using correlations. These include

- Spurious correlation
- Outliers
- Reversing cause and effect
- Confounders and Mediators

Spurious Correlation

You may have heard the famous maxim *"correlation does not imply causation"*. Indeed, having a strong correlation alone may be much less informative than you think. The fact that two variables X and Y are correlated may be attributed to

i. X influencing Y (in which case we say X has a *causal effect* on Y), or vice versa,
ii. both X and Y being influenced by another variable Z (which is called a confounder),
iii. pure coincidence.

Although in applied statistics, the question we often want to answer is whether the statistically significant correlations we observed is due to *(i)*, unfortunately in most cases it is a result of *(ii)* or *(iii)*. You can find many datasets that are correlated but have no causal effect on one another. For example,

- *Per capita consumption of mozzarella cheese* correlates with *civil engineering doctorates awarded* (in the US - see this website (http://tylervigen.com/spurious-correlations#:~:text=Cheese%20consumed-,Per,-capita%20cheese%20consumption)),
- *Ice cream consumption in the US* correlates with its *crime rate*,
- *A ranking of countries based on confirmed COVID-19 cases* correlates with *FIFA ranking* (Ayoub et al., 2020 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7470813/)).

Exercise

1. The first website linked above does a good job of showing some interesting spurious correlations. What are your thoughts on the visualisations used to illustrate these correlations? Is there anything you would change?

These are examples of spurious correlation, which occur when two or more variables are correlated but *not* causally related (i.e. one does not cause the other, directly or indirectly). Spurious correlation can occur due to counfounding, or simply due to coincidence. In large datasets, however, the occurrence of 'coincidental' spurious correlation in subsets of the data can be a statistically likely phenomenon; we illustrate this with the following synthetic example.

Example: Normal Random Variables

Exercises

Follow the following steps to draw 10000 groups of samples $\{(x_i, y_i)\ ;\ i = 1, \ldots, 25\}$, where x_i, y_i are i.i.d. Gaussian random variates. We will analyse the correlation between x_i and y_i in each group.

2. Create a vector of group indices of the form $(1, 1, \ldots, 1, 2, 2, \ldots, 2, \ldots, g, g, \ldots, g)$, where $g = 10000$ is the number of groups, and each index is repeated $N = 25$ times. Assign it to the variable `group`.
3. Draw $N * g$ realisations from $\mathcal{N}(0, 1)$ and assign it to the variable `x`.
4. Repeat the previous step for the variable `y` and place these three variables in the tibble `normal_data`.

```
N <- 25
g <- 10000

group <- rep(1:g, each=N) #Q2
x <- rnorm(N * g) # Q3
y <- rnorm(N * g) # Q4

normal_data <- tibble(group = group,
                      x = x,
                      y = y) #Q4
```

5. Use `group_by` to group the dataframe by `group`.
6. Use `summarize` and `cor()` to compute the correlation `r` *within* each group.
7. Use `arrange` to sort the dataframe in decreasing order of `r` (*hint: use the built-in function* `desc()`). Call the resulting tibble `sorted_data`.

```
sorted_data <- normal_data %>%
  group_by(group) %>% # Q5
  summarize(r = cor(x, y)) %>% # Q6
  arrange(desc(r)) # Q7

head(sorted_data)
```

```
## # A tibble: 6 × 2
##   group     r
##   <int> <dbl>
## 1  7426 0.673
## 2  2933 0.648
## 3  7728 0.643
## 4  1368 0.638
## 5  2704 0.627
## 6  7923 0.602
```

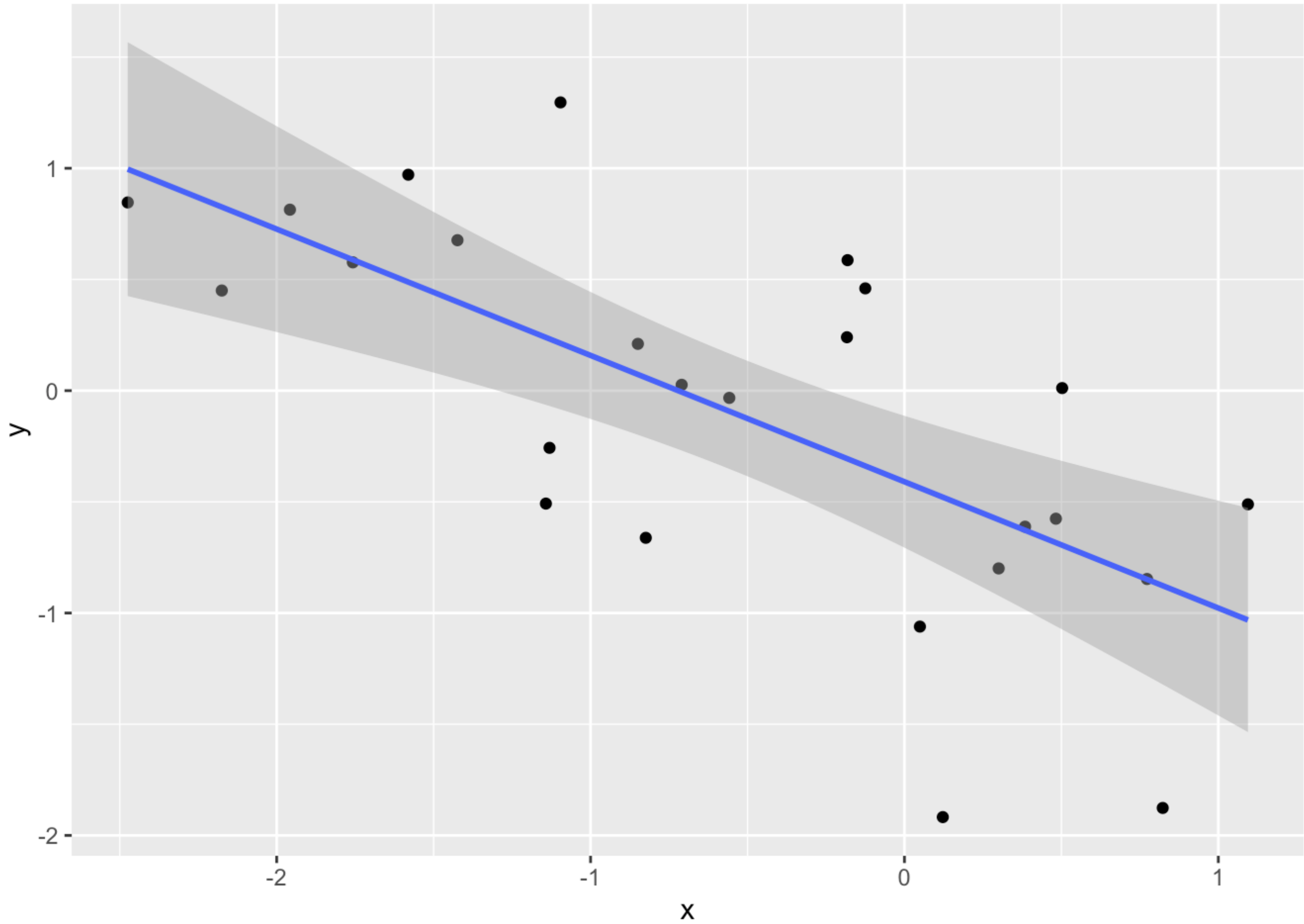
We now find the group that has the smallest correlation, and plot the data pairs in that group.

Exercises

8. Use `which.min()` to find the index corresponding to the group with the smallest value of `sorted_data$r`.
9. Use a scatter plot of x_i against y_i to visualise the data in this group. Fit a linear model to this data. What can you conclude about their correlation? Is your observation surprising?

```
index_min <- which.min(sorted_data$r) # Q8

# Q9:
# we filter sorted_data, using the index found above. This returns the group of da
ta with the lowest correlation coefficient
neg_corr_data <- normal_data %>%
  filter(group == sorted_data$group[index_min])

# We use geom_point to construct a scatter plot of the data in this group.
# As we are going to fit a linear model (below), we can also pre-emptively use geo
m_smooth() to illustrate the results of this model
neg_corr_data %>%
  ggplot(aes(x, y)) +
  geom_point() +
  geom_smooth(method = "lm") # draw the best-fitted line with OLS
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



In the following cell, we fit a linear model to this group of data; we see that the fitted coefficient for X is negative and is statistically significant. However, by construction, we known that X and Y are independent, and so this observed correlation must be spurious!

```
lm_fit <- lm(y~x, data=neg_corr_data)
summary(lm_fit)
```

```
##
## Call:
## lm(formula = y ~ x, data = neg_corr_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.43856 -0.37561  0.03261  0.48277  1.08327
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.4096     0.1432  -2.859  0.00887 **
## x            -0.5680     0.1274  -4.458  0.00018 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6263 on 23 degrees of freedom
## Multiple R-squared:  0.4635, Adjusted R-squared:  0.4402
## F-statistic: 19.87 on 1 and 23 DF,  p-value: 0.0001798
```

The above exercises demonstrate a particular instance of data dredging known as cherry-picking. It refers to the action of looking through many results produced by a random process and picking the one that shows a relationship that supports a theory you want to defend.

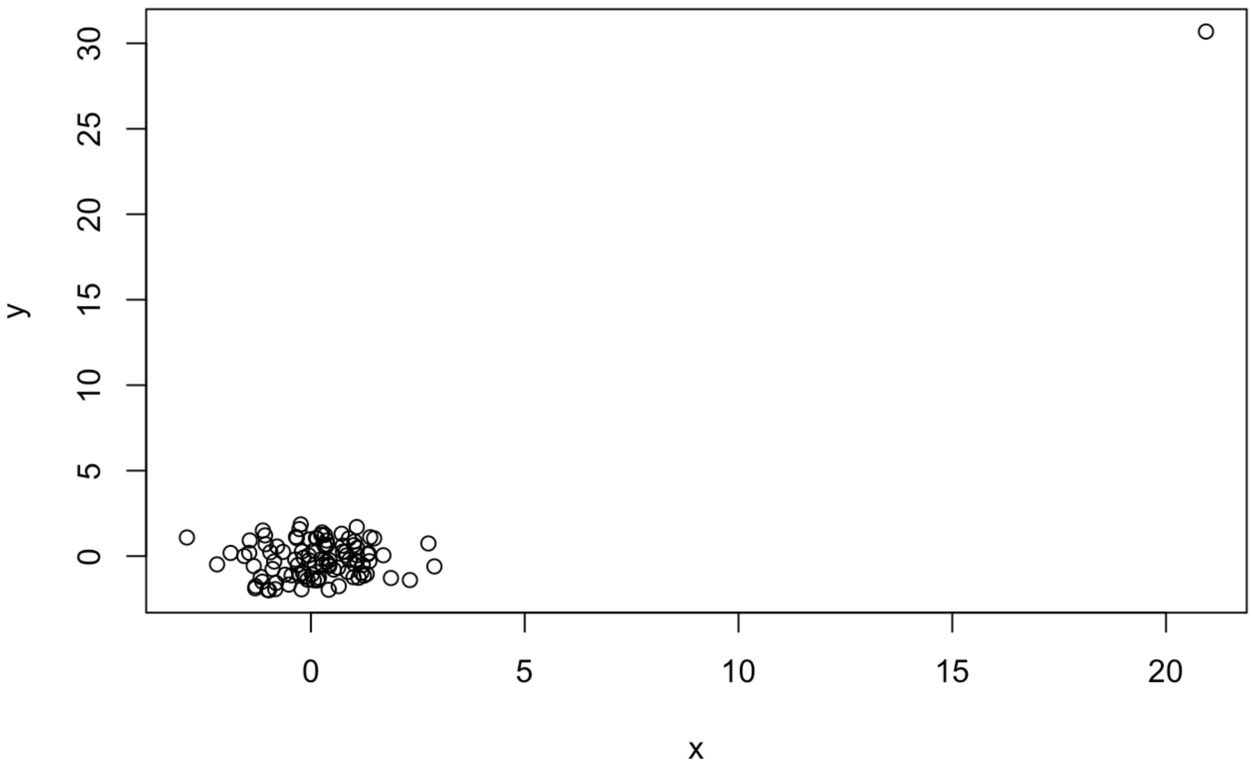
In particular, we can see from this example that as long as we can generate sufficiently many pairs of random variates, we can always find a subset where the correlation between the pair is statistically significant. This particular form of data dredging is also known as p-hacking.

Rather surprisingly, p-hacking is a common mistake researchers make when e.g. including in their publications only those significant results that would support their claims. This could be due to unethical behaviour where researchers deliberately present biased findings to make their papers more likely to be published, but it is also often a result of statistical ignorance.

Outliers

Correlation can also be affected by the presence of outliers. As an example, we simulate independently 100 pairs of data points $\{(x_i, y_i)\}_{i=1}^{100}$, where the first 99 pairs are drawn from a standard Gaussian distribution, and the last pair are such that $x_{100} \sim \mathcal{N}(20, 4)$ and $y_{100} \sim \mathcal{N}(30, 4)$.

```
set.seed(2022)

x <- rnorm(100)
y <- rnorm(100)
x[100] <- x[100] * 2 + 20
y[100] <- y[100] * 2 + 30

plot(x, y)
```



Exercises

10. Use `cor()` to compute the correlation between `x` and `y` .
11. Repeat the above computation, but this time without the 100-th elements of `x` and `y` . What do you find?

```
corr_all <- cor(x, y) #Q
corr_sub <- cor(x[-100], y[-100]) #Q

cat(sprintf("correlation for all data\t\t: %.2f \ncorrelation excluding the 100th
datum\t: %.2f",
        corr_all,
        corr_sub))
```

```
## correlation for all data     : 0.86
## correlation excluding the 100th datum    : 0.06
```

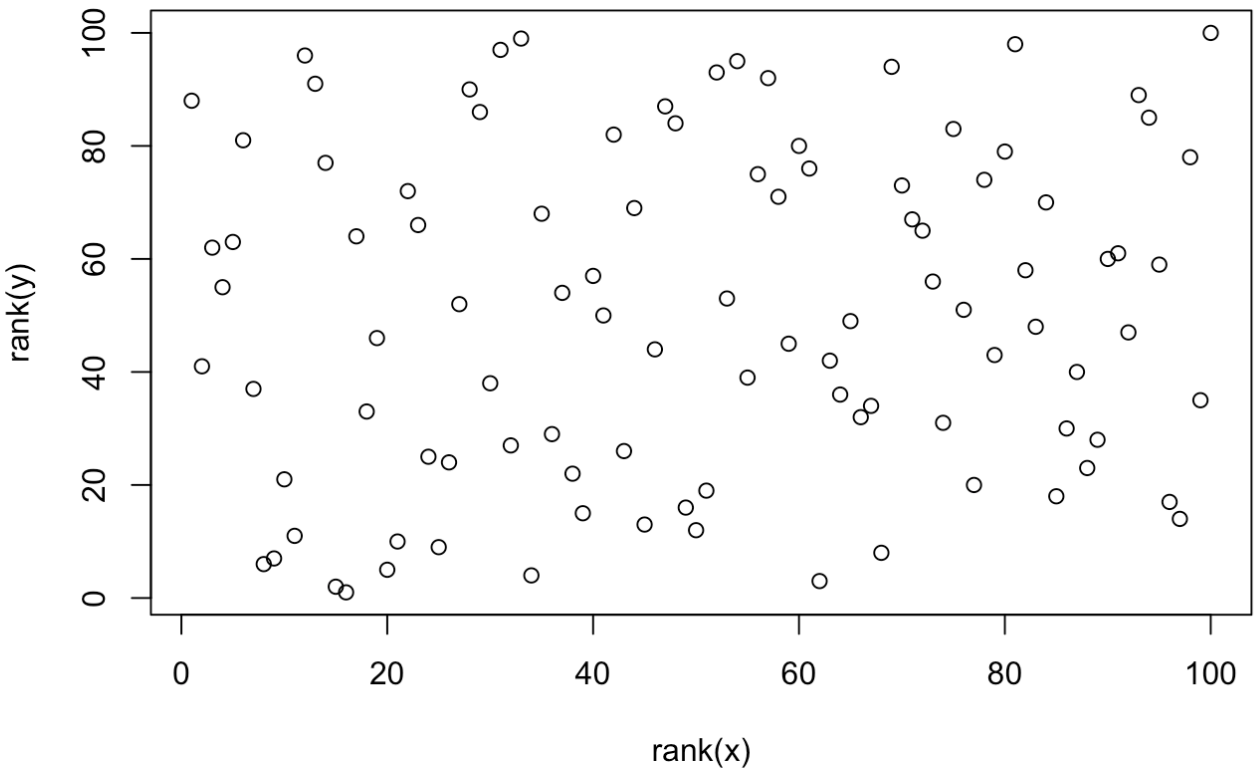
From these exercises, you should see that adding the outlier has increased the correlation from almost 0 to almost 1. This is essentially a result of the fact that the sample mean and sample variance are sensitive to outliers (recall from Week 3). This can be understood with some simple analysis: if we have an observed sample $(x_1, x_2, \ldots, x_n)$ where $|x_n| \gg \max_{i=1,\ldots,n-1}(|x_i|)$, then the sample mean is approximately $\overline{x} \approx \frac{1}{n} x_n$. Repeating this analysis for the second moment of $(x_1, x_2, \ldots, x_n)$ and the second joint moment between $(x_1, x_2, \ldots, x_n)$ and another sample $(x_1, x_2, \ldots, y_n)$, we can show that the correlation between these two samples is $\mathrm{corr}(\{x_i\}_i^n, \{y_i\}_i^n) \approx 1$.

Spearman's Correlation Coefficient

The above example shows that Pearson's correlation coefficient can be deceptive in the presence of outliers. One possible solution is to *standardise* your data before any analysis and work with $\{(x_i - \overline{x})/s\}_i^n$, where $\overline{x}$ is the sample mean and s is the sample standard deviation. Another solution is to use a correlation coefficient that is *robust* to outliers (as discussed in Week 3). One such robust coefficient is Spearman's correlation coefficient, which simply computes the correlation on the ranks of the values.

By constructing a scatterplot of `rank(y)` against `rank(x)`, we can see that the outlier is no longer associated with a large value.

```
plot(rank(x), rank(y))
```



Spearman's correlation coefficient exploits this loss of information to provide a measure of the correlation that is ignorant to the actual values of the dataset. We can calculate this straightforwardly as follows:

```
cor(rank(x), rank(y)) # or equivalently cor(x, y, method = "spearman")
```

```
## [1] 0.1067867
```

Reversing Cause and Effect

Another form of confusion between correlation and causation arises when the dependent and independent variables are reversed in the process of determining the causal relationship. Consider the following example:

> A statistician observes a positive correlation between the amount of coffee consumed by a group of statistics students and their levels of anxiety. The statistician theorizes that drinking coffee causes nervousness, but instead finds that nervous people are more likely to drink coffee.

It is easy to find two variables for which a strong correlation remains even when the cause and the effect are reversed. Consider the `cars` dataset, which gives the speed of cars and the distances taken to stop.

```
head(cars)
```

```
##   speed dist
## 1     4    2
## 2     4   10
## 3     7    4
## 4     7   22
## 5     8   16
## 6     9   10
```

We can fit the following linear model:

$$X_i = \beta_0 + \beta_1 Y_i + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2),\ i = 1, \ldots, 50,$$

where X_i denotes the speed and Y_i denotes the distance.

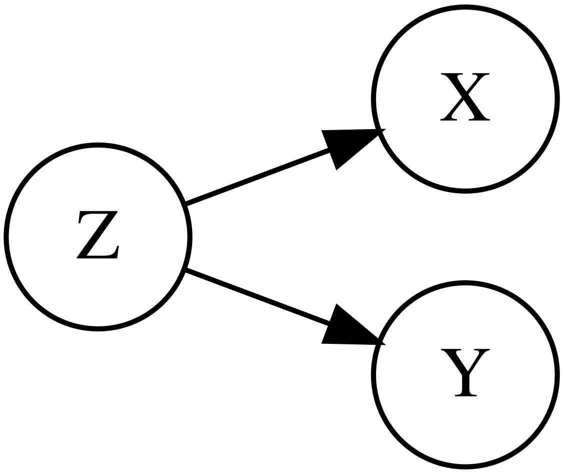
```
lm_fit <- lm("speed~dist", data=cars)
summary(lm_fit)
```

```
##
## Call:
## lm(formula = "speed~dist", data = cars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.5293 -2.1550  0.3615  2.4377  6.4179
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.28391    0.87438   9.474 1.44e-12 ***
## dist         0.16557    0.01749   9.464 1.49e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.156 on 48 degrees of freedom
## Multiple R-squared:  0.6511, Adjusted R-squared:  0.6438
## F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12
```

Clearly, β_1 is statistically significant. Had we overlooked the context, it is easy to conclude falsely that *"Y has a significant causal effect on X"*. Note that the model is technically correct; what is wrong is the interpretation. A careful statistician should always interpret the results of their analysis based on the context of the dataset.

Confounders and Mediators

Given two correlated variables X and Y, another variable Z is a confounder if changes in Z causes changes in both X and Y. This relationship is depicted by the following figure, where an arrow indicates a causal effect:



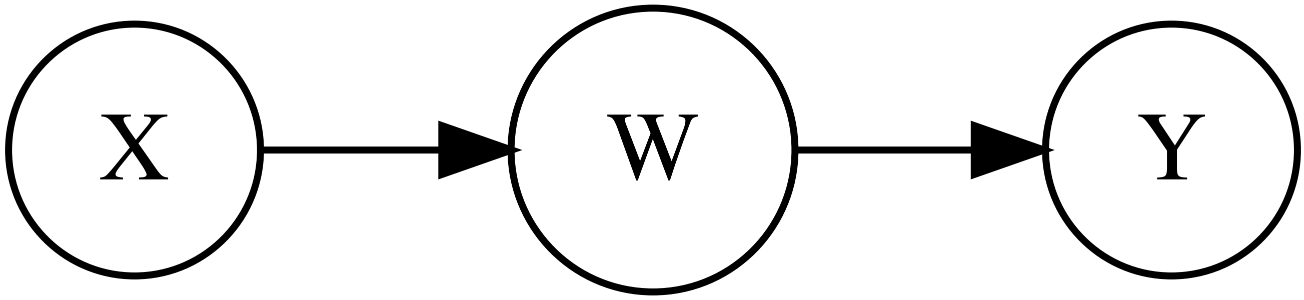
Confounders can be measured or unmeasured; this refers to whether or not the confounder is observed in the study (and hence can be conditioned upon in an analysis). Consider the following example:

> On any given day, an individual observes whether the grass on the Queen's Lawn at Imperial College is wet, and they also observe whether the majority of pedestrians on campus are carrying an umbrella. They note a significant correlation between the two: typically, when the Queen's Lawn is wet, the majority of pedestrians carry umbrellas. The individual concludes that the wet grass causes pedestrians to carry an umbrella.

This is a toy example in which two events are correlated due to an obvious confounding variable. The wet grass will not *cause* an increase in pedestrians carrying umbrellas, and similarly vice versa. Instead, it is much more likely that both are the result of rainy weather. In this case, whether it rains is the confounder.

Failing to properly account for the true causal relationships between variables in a multivariate dataset can result in misleading analyses. This is often seen when confounding variables are ignored, however the risk is not limited to the presence of confounders. In some applications, the dependence between two variables can be shown to change dramatically when conditioning on the value of another variable, known as a *mediator*.

Given two correlated variables X and Y, another variable W is a mediator if a direct causal relationship between X and Y can be explained by two causal relationships: X influences W and W influences Y.. This relationship is depicted by the following figure:



Simpson's Paradox

We will conclude this notebook with two examples, one simulated and another based on a real study, which demonstrate *Simpson's paradox*; this is where the dependence between two variables changes when an additional variable is accounted for. Simpson's paradox can be observed in the presence of confounding variables, however the two examples that follow both consider the existence of an additional mediator variable.

Simulated Example

Suppose we are interested in analysing a potential link between the annual salary of a set of individuals and the annual income of their parents. Let us assume the following model:

$$\mathtt{salary} = 30 + 30 * \mathtt{education} - 0.4 * \mathtt{parent} + 15 * \epsilon,$$

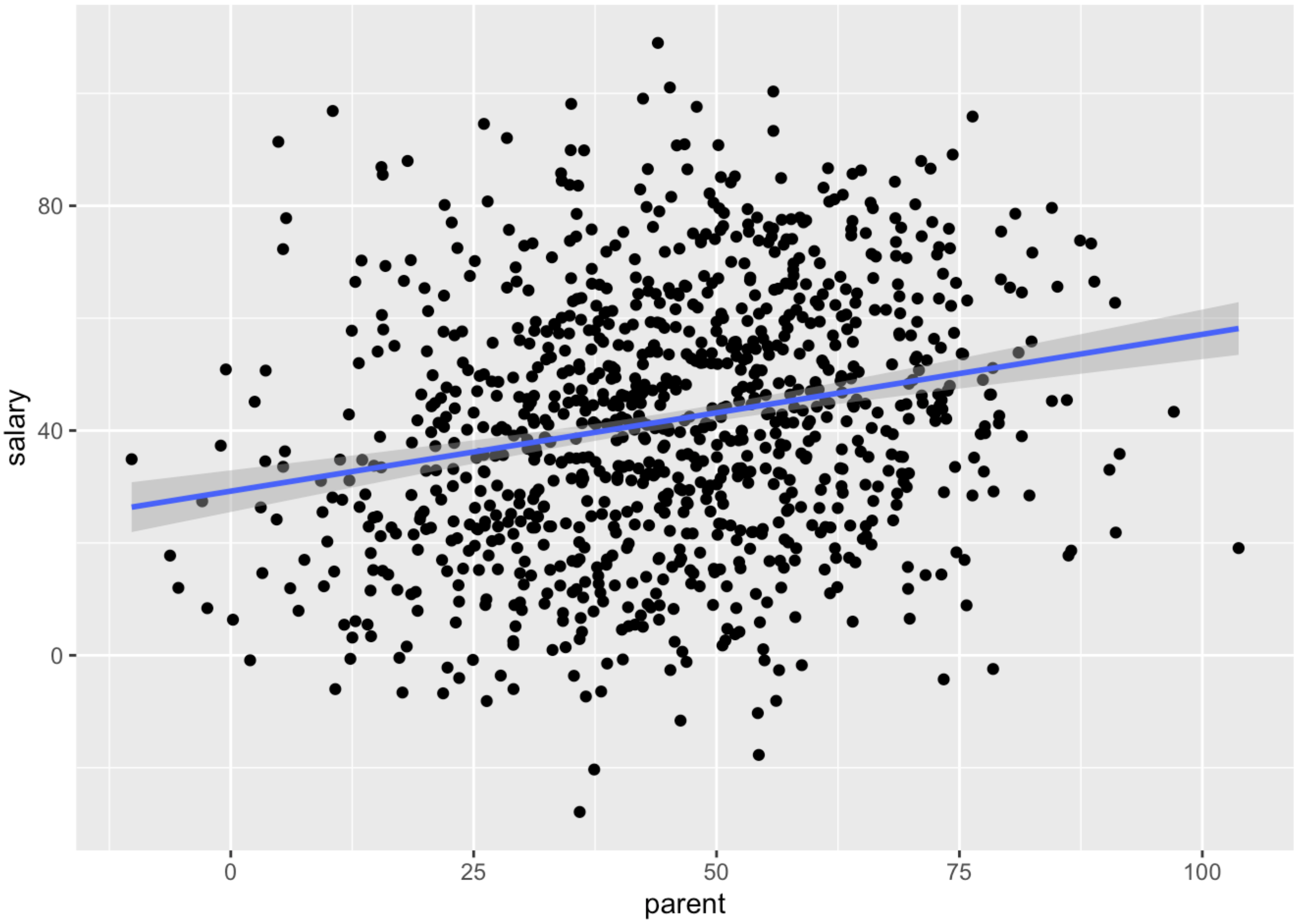
where $\mathtt{salary}$ is the annual salary of the subject (in '000s GBP), $\mathtt{education} \in \{0, 1, 2\}$ is a categorical variable recording the level of formal education undertaken by them, $\mathtt{parent} = 30 + 15 * (\mathtt{education} + \delta)$ is the annual income of their parents (in '000s GBP), and ϵ, δ are i.i.d. noise terms that follow a standard Gaussian distribution. Note in particular that $\mathtt{education}$ mediates the relationship between $\mathtt{parent}$ and $\mathtt{salary}$ in this model. We generate 1000 data points in `R`:

```
set.seed(2022)
n = 1000

education <- rbinom(n, 2, 0.5)
parent <- 30 + 15*(education + rnorm(n))
salary <- 30 + 30*education - 0.4*parent + 15*rnorm(n)

education <- factor(education, labels = c("Low", "Medium", "High"))

data <- data.frame(
  education,
  salary,
  parent
)
```

We now plot the subjects' annual salary against their parents income for the entire data. We can see a clear upward trend, suggesting that annual income is positively correlated with parental income.

```
p <- data %>% ggplot(aes(parent, salary))
p + geom_point() +
  geom_smooth(method = 'lm')
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

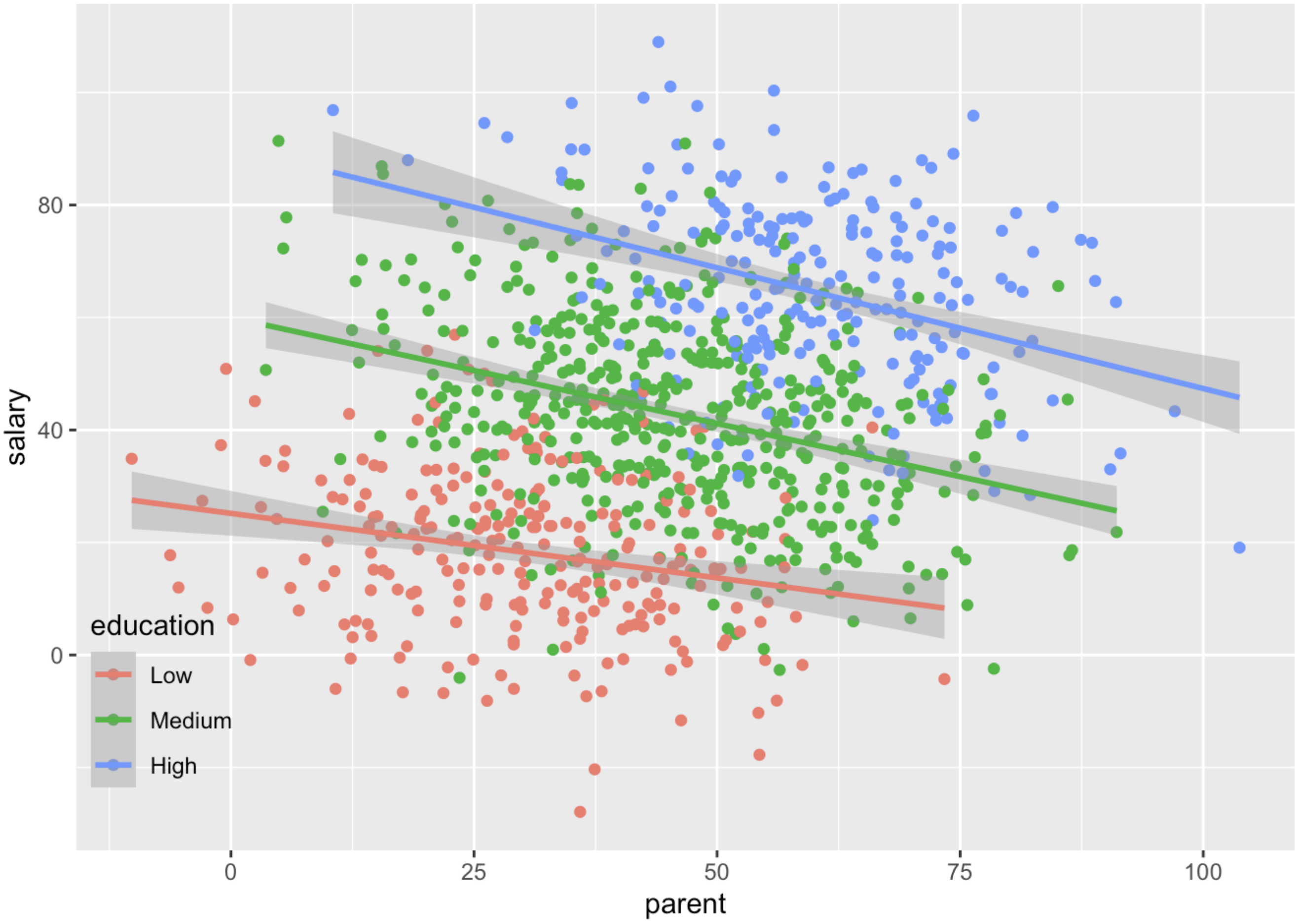


However, if we divide the data into groups by the level of education, and plot the data separately for each group, we see that the trend would be reversed.

```
p + geom_point(aes(col = education)) +
    geom_smooth(aes(col = education), method = 'lm') +
    theme(legend.background = element_rect(fill = "transparent"),
          legend.justification = c(0, 1),
          legend.position = c(0, 0.3))
```

```
## Warning: A numeric `legend.position` argument in `theme()` was deprecated in gg
plot2
## 3.5.0.
## i Please use the `legend.position.inside` argument of `theme()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Hence, the conclusion drawn on the association between two variables is qualitatively different after controlling on another variable.

Example: UC Berkeley admissions

We look at a dataset of admission statistics from six U.C. Berkeley majors from 1973; the following is a classic example of Simpson's paradox at work. The data can be loaded as a 3-dimensional table `UCBAdmissions` in `R`. Here, `Dept` refers to the major to which the students applied.

```
data(UCBAdmissions)
class(UCBAdmissions) # this dataset is a table
```

```
## [1] "table"
```

```
dim(UCBAdmissions) # dim of the table
```

```
## [1] 2 2 6
```

```
dimnames(UCBAdmissions) # name and classes of each dimension
```

```
## $Admit
## [1] "Admitted" "Rejected"
##
## $Gender
## [1] "Male"   "Female"
##
## $Dept
## [1] "A" "B" "C" "D" "E" "F"
```

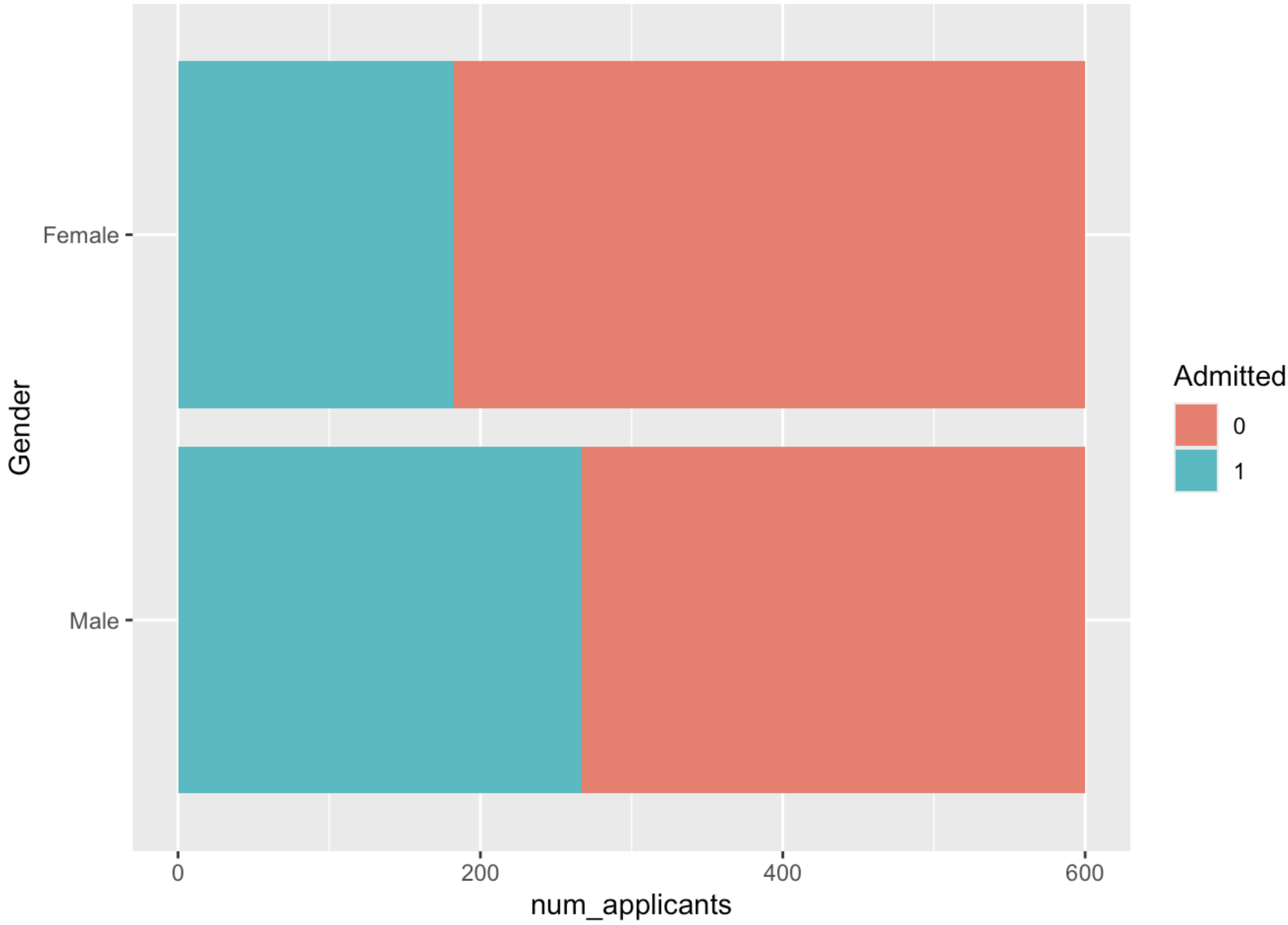
We first convert the 3D table into a 2D dataframe.

```
df_admitted <- UCBAdmissions["Admitted", ,] %>%
  data.frame() %>%
  mutate(Admitted = 1)

df_rejected <- UCBAdmissions["Rejected", ,] %>%
  data.frame() %>%
  mutate(Admitted = 0)

admissions_df <- rbind(df_admitted, df_rejected) %>%
  mutate(Admitted = factor(Admitted)) # change datatype to categorical

head(admissions_df)
```

```
##   Gender Dept Freq Admitted
## 1   Male    A  512        1
## 2 Female    A   89        1
## 3   Male    B  353        1
## 4 Female    B   17        1
## 5   Male    C  120        1
## 6 Female    C  202        1
```

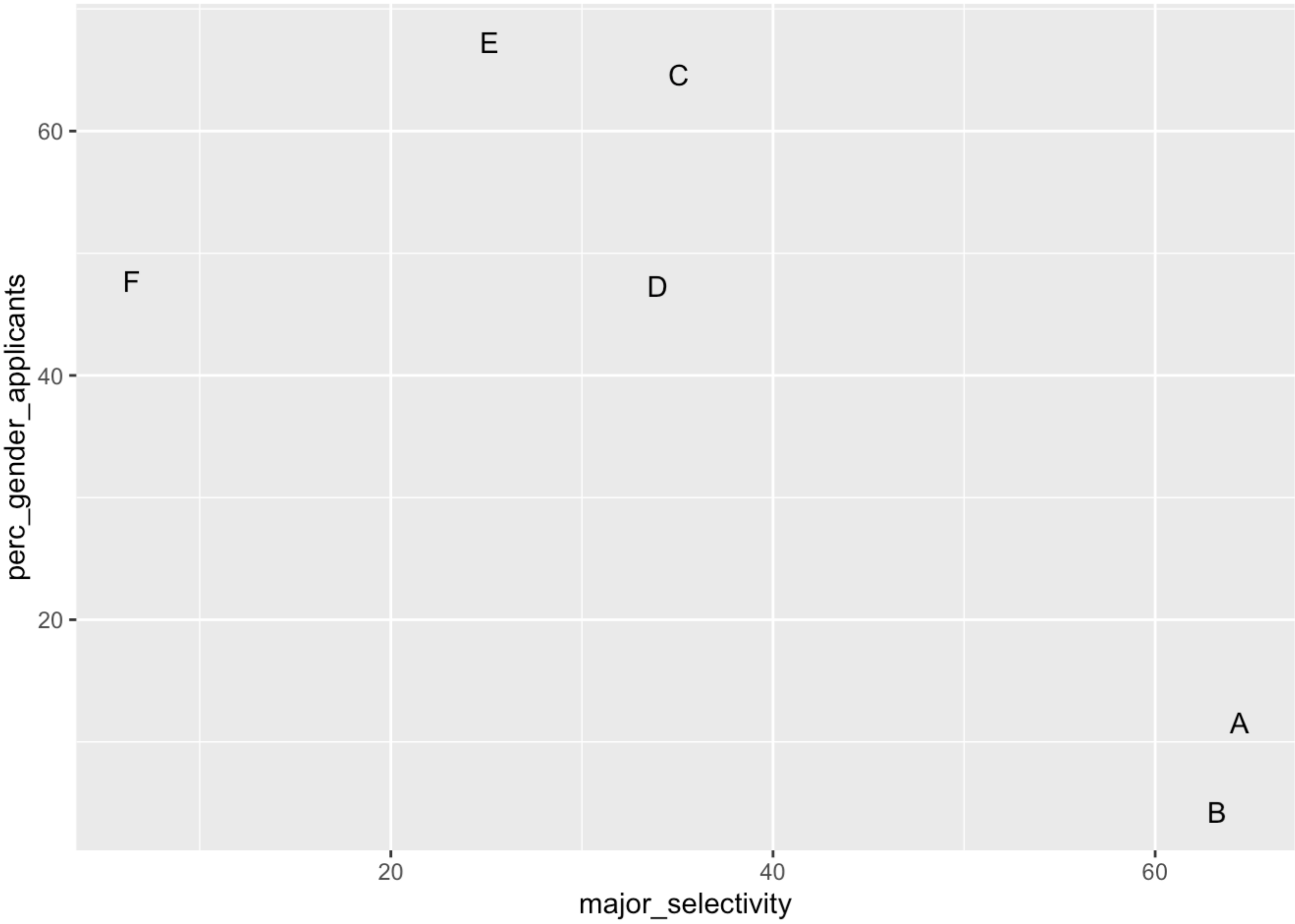
If we plot the admission rates by gender, we clearly see that more females were rejected than males, so it seems to suggest female applicants were disadvantaged in the application process.

```
admissions_df %>%
  group_by(Gender) %>%
  mutate(gender_applicants = sum(Freq)) %>%
  group_by(Gender, Admitted) %>%
  mutate(num_applicants = sum(Freq) / gender_applicants * 100) %>%
  ggplot(aes(x = num_applicants,
             y = Gender)) +
  geom_col(aes(fill = Admitted))
```



However, a closer inspection shows a different result. Below we compute the admission rates *by major*.

```
admissions_df <- admissions_df %>%
  data.frame() %>%
  group_by(Dept) %>%
  mutate(applicants = sum(Freq)) %>%
  group_by(Dept, Admitted) %>%
  mutate(major_selectivity = 100 * sum(Freq) / applicants) %>%
  group_by(Dept, Gender) %>%
  mutate(perc_gender_applicants = 100 * sum(Freq) / applicants)

admissions_df %>%
  filter(Gender == "Female",
         Admitted == 1) %>%
  ggplot(aes(major_selectivity, perc_gender_applicants, label = Dept)) +
  geom_text()
```



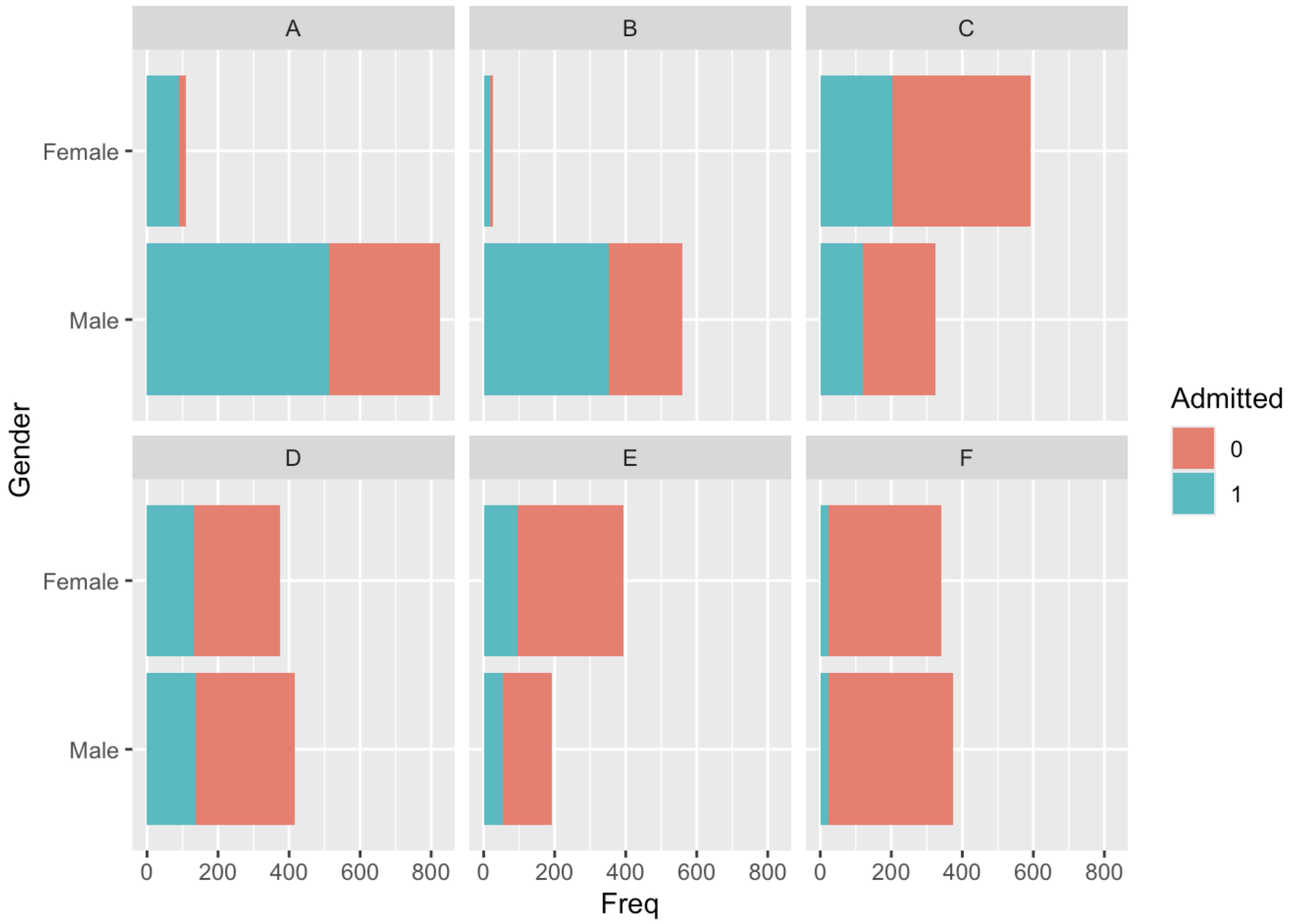
We see that the majors with a low admission rate received more female applicants than males. For example, over 60% of applicants to major E were female and this major had an admission rate of about 25%; in comparison, for major A for which only roughly 12% applicants were female, the admission rate was over 60%.

We then plot the number of admitted applicants by gender and major.

Exercise

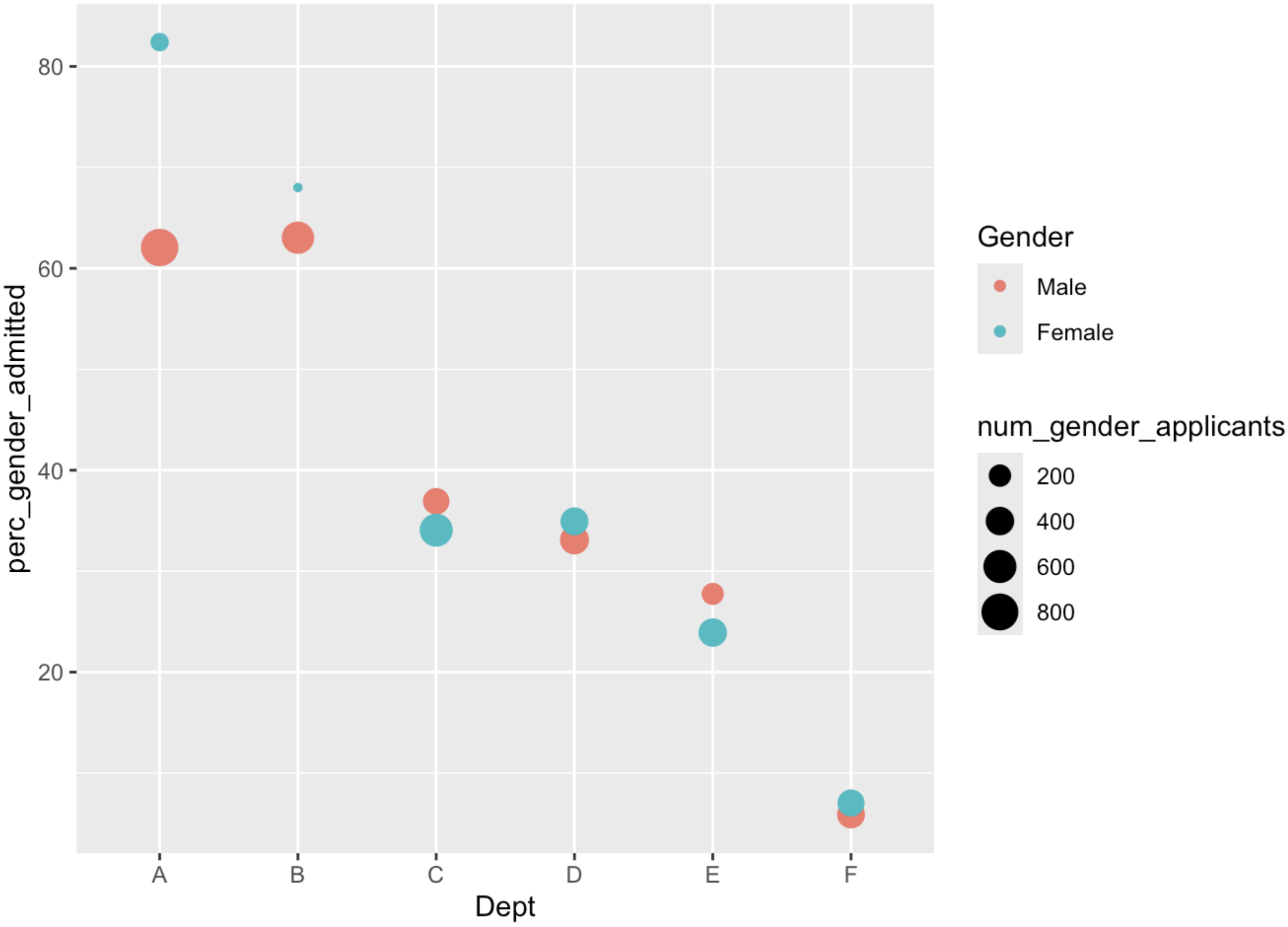
12. Complete the code below to plot a bar chart of the number of admitted applicants (`Freq`) by gender (`Gender`).

```
admissions_df %>%
  ggplot(aes(x = Freq,
             y = Gender)) +
  geom_col(aes(fill = Admitted)) +
  facet_wrap(~Dept)
```



The resulting plot should show that the majority of accepted males come from two majors: A and B. Fewer females applied to these two majors, and hence less females were admitted. This motivates us to compute the difference in the admission rates *conditioned on the major*.

```
admission_rates_df <- admissions_df %>%
  filter(Admitted == 1) %>%
  mutate(num_gender_applicants = (perc_gender_applicants/100 * applicants),
         perc_gender_admitted = 100 * Freq / num_gender_applicants)

admission_rates_df %>%
  ggplot(aes(Dept, perc_gender_admitted, col = Gender, size = num_gender_applicant
s)) +
  geom_point()
```



This plot suggests that, within each major, there is little difference between the admission rates of male and female applications. Here, the size of the dots are proportional to the number of applicants. We see in particular that even though majors A and B had many more male applicants, the admission rate is actually higher amongst females in these two groups.

Exercises

Finally, we compute the mean difference of the admission rates by major by

13. Using `group_by` to group the data by `Gender`.
14. Setting a new variable `average` in `summarise()` to be the mean of `perc_gender_admitted` to compute the mean admission rates for each major.
15. What do you conclude?

```
admission_rates_df %>%
  group_by(Gender) %>%
  summarise(average = mean(perc_gender_admitted))
```

```
## # A tibble: 2 × 2
##   Gender average
##   <fct>    <dbl>
## 1 Male      38.1
## 2 Female    41.7
```

We find that, when conditioning on the department/major applied to, the admissions rate for females is actually higher than the admissions rate for males, with a difference of 3.6%.